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Are Healthy Foods Affordable? The Past, Present, and Future of Measuring Food Access Using Least-Cost Diets

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Abstract

This review describes the history, current practice, and prospects for measuring a population's access to foods for health using the lowest-cost locally available items, in contrast to quantities actually chosen, so as to distinguish between unaffordability of healthy diets and other causes of malnutrition. Retail prices, cost per day, and affordability relative to earnings have been used to measure food access for centuries, driving early definitions of poverty and income thresholds for subsistence. Substitution between items based on food composition was introduced soon after nutrient requirements were quantified, leading to the development of linear programming and other diet modeling techniques. This article describes how and why modern diet cost and affordability metrics have been adopted by international organizations, government agencies, and researchers to monitor food markets and guide intervention, concluding with new frontiers for research on other factors limiting access to healthy diets, such as cooking costs and time use, and factors that cause displacement of low-cost healthy diets by other foods.

1. INTRODUCTION AND MOTIVATION

Having sufficient food to meet nutritional requirements is one of humanity's oldest concerns, with documented records dating as far back as 5,000 years ago in Mesopotamia (Postgate 1994). Mesopotamian records include cereal rations designed to meet the basic requirements of workers, accompanied by specification of more diverse, ideal diets to which higher-income people could aspire (Ellison 1981). In the modern era, monitoring the adequacy of the global food supplies has been a central focus of international cooperation for over a century, since the League of Nations (1920) began its *Monthly Bulletin of Statistics* tracking changes in food prices and quantities around the world, leading to modern systems of monitoring major farm commodities entering international trade (FAO 2023a, 2024c; World Bank 2023a).

Recent innovations in the measurement of food access have been driven by both policy priorities and methodological advances in data collection and analysis. The current internationally accepted definition of food security, established at the World Food Summit of 1996, is "when all people at all times have physical and economic access to sufficient, safe and nutritious food to meet their dietary needs and food preferences for an active and healthy life" (Comm. World Food Secur. 2012). Different aspects of that concept have been measured in different ways, most recently by defining access as affordability of the lowest-cost locally available items in quantities needed to meet national governments' food-based dietary guidelines. These diet cost and affordability metrics are designed to distinguish access from utilization and thereby guide change in food environments, as discussed by Herforth & Ahmed (2015), in addition to the previously available set of indicators described by Lele et al. (2016).

This review summarizes the analytical methods, data sources, and policy applications for the least-cost diets introduced in the late 2010s and now used to monitor food access globally. The purpose of these least-cost diets is to provide a benchmark that differs from actual consumption in ways that help distinguish among causes of poor diet quality, and thereby guide three types of intervention: (*a*) Where and when least-cost items are unusually expensive, access to healthy diets can be improved by investments in agricultural production and food distribution toward the frontier of lowest costs observed elsewhere; (*b*) if prices are at the low-cost frontier but available incomes are even lower, healthy diets remain unaffordable, so improving the nutritional quality of diets would require increased earnings, lower nonfood expenditure, or social assistance and safety nets; and (*c*) if a low-cost healthy diet would be affordable but those foods are displaced by other items, remedies would need to address drivers of food choice beyond price and income, such as preferences and aspirations shaped by marketing, or time use and cooking costs in meal preparation.

Calculating least-cost diets that meet nutrient requirements was initially attempted by Stigler (1945) and was done primarily to formulate rations for the military and other feeding programs. Using least-cost items by food group to track international variation in a global standard for diet diversity was introduced by Masters et al. (2018) in a study of monthly fluctuations by region in Ghana and Tanzania and by Herforth et al. (2020) for global measurement of costs per day and the total number of people unable to afford healthy diets as a background paper for the United Nations (UN) agencies' flagship report on *The State of Food Security and Nutrition in the World* (FAO et al. 2020). Subsequent updates were designed to facilitate monitoring of changes from year to year, as detailed in Herforth et al. (2022), including especially the formulation of food group targets for a Healthy Diet Basket (HDB) (Herforth et al. 2025) used in the suite of metrics known as the Cost and Affordability of a Healthy Diet (CoAHD) now calculated annually by international agencies (FAO 2024a, World Bank 2024b) and national governments such as Nigeria's National Bureau of Statistics (Natl. Bur. Stat. 2024).

We begin this review with a brief summary of food price and quantity data available prior to the recent work on cost and affordability of healthy diets. We then describe how least-cost diets based on nutrient requirements were developed and used to guide nutrition assistance in the United States and globally, and how monitoring least-cost healthy diets based on food groups can allow decision-makers to distinguish access from use. This in turn guides efforts to end malnutrition by improving food supply toward the frontier of lowest cost, raising incomes to make those diets affordable or limiting displacement of healthy diets by less nutritious foods, as described by Popkin (1993), Drewnowski & Popkin (1997), and others.

2. PRICES, INCOMES, AND PREVIOUS APPROACHES TO FOOD SECURITY MEASUREMENT

Food prices relative to incomes are among the first societal conditions to have been measured, with patterns of change clearly revealed by data for recent decades. An early English-language description of how a diet cost metric could be used to assess the adequacy of wages was written nearly 350 years ago by Rice Vaughan (1675). Despite the archaic wording, modern readers will recognize the idea that wages paid to workers should account for the cost of sufficient food, clothing, and other things needed for daily subsistence:

> The hire of Labourers and Servants carrieth with it a resultance of the prices of all things generally necessary for a mans life: Besides, that Reason doth convince that there must be a convenient Proportion between their Wages and their Food and Raiment, the Wisdom of the Statute doth confirm it, which doth always direct the Rate of Labourers and Servants to be made with a regard of Prices of Victuals, Apparel, and other things necessary to their use.

The first recorded effort in English to actually compute affordability is due to William Fleetwood (1707), prompted not by the needs of low-wage workers but by student requests for more generous stipends at University of Oxford. The income threshold for fellowships had been set at £5 per year more than 150 years earlier, and Chance (1966, p. 109) quotes Fleetwood's explanation of the need for an increased allowance, which spells out much of the logic behind modern food price indexes:

> To know somewhat more distinctly whereabouts an Equivalent to your ancient £5 will come, you are (as I before hinted) to observe how much Corn, Meat, Drink, or Cloth, might have been purchased 250 Years ago, with £5 and to see how much of the modern Money will be requisite to purchase, the same quantity of Corn, Meat, Drink, or Cloth, now-a-days. To this End, you must neither take a very dear Year, to your Prejudice, nor a very cheap one, in your own Favor, nor indeed any single Year, to be your Rule; but you must take the Price of every particular Commodity, for as many Years as you can (20, if you have them) and put them all together; and then find out the common Price; and afterwards take the same Course with the Price of Things, for these last 20 years, and see what Proportion they will bear to one another; for that Proportion is to be your Rule and Guide.

Fleetwood's analysis of students' cost of living was based on a fixed basket with 5 quarters of wheat, 4 hogsheads of beer, and 6 yards of cloth. A variety of similar consumption baskets were later proposed for different purposes, such as the weighted average of market prices used by the Massachusetts State legislature in 1780 to adjust the pay of Revolutionary War soldiers, based on a basket of 5 bushels of corn, 68.6 pounds of beef, 10 pounds of sheep's wool, and 16 pounds of sole leather (Diewert 1988).

The concept of affordability links food prices and diet costs to a measure of earnings, the analysis of which dates at least to William Playfair (1821), who charted the price of bread relative to workers' wages in England. Playfair's analysis showed how bread had become increasingly affordable in the seventeenth and early-eighteenth centuries, but that this history of improvement ended when bread prices began to rise at roughly the same rate as wages. The title of Playfair's pamphlet, "A Letter on Our Agricultural Distresses, Their Causes and Remedies," summarizes how measuring the affordability of food could guide action by comparing food prices to available income. Playfair's chart of the decline followed by stability in the real cost of bread marks the earliest well-documented analysis of diet costs and affordability, using only the one type of food and one source of income for which data were available (**Supplemental Figure 2**).

Supplemental Material >

Over the nineteenth and twentieth centuries people began to consume an increasingly wide range of foods and other things, sparking interest in cost-of-living indexes that would reflect the total cost of diverse items, weighting each price by its share of expenditure to track the average cost of what people buy. Étienne Laspeyres (1871) advocated for using quantities consumed during the initial observation period, while Hermann Paasche (1874) argued for using the ending period quantities. If the purpose of price indexes is to track the cost of achieving a specific living standard, then indexes should allow for a population's willingness to substitute between items as prices change. In the twentieth century, analysis of household survey data allowed those substitution possibilities to be represented by empirically estimated demand systems, summarizing a population's choices in response to income and price changes using a set of simultaneous equations whose functional forms and parameter restrictions are internally consistent (Deaton & Muellbauer 1980). This led to the development of chained price indexes, whose weights vary over time to match each population's expenditure share for each kind of item in the index (Diewert et al. 2009).

The availability of comprehensive price indexes over many goods and services allows the cost of each item to be expressed in real terms, relative to the prices of all other commodities. Some ongoing work is pursuing Playfair's comparison of food prices and diet costs to laborer's wages (Headey et al. 2024a), but most analyses focus on comparing observed nominal prices of food in each current period to a consumer price index (CPI) to reflect the real purchasing power of money in a constant base period. The resulting CPI is now calculated monthly by almost every national government in the world, with global efforts to harmonize methods and reporting procedures led by the UN Statistical Commission and other international agencies (Graf 2020).

Comparing food environments in terms of access to a given dietary standard must contend with the diversity of income sources, especially because farmers might benefit from higher prices for what they sell (e.g., Headey 2018), prompting the use of total earnings instead of wages. Analysts must also contend with the diversity of items consumed, leading to the use of food price indexes instead of individual item prices, and the role of nonfood goods and services, driving the use of a general CPI to adjust for inflation. It is also important to account for spatial differences in price levels, calling for adjustments based on purchasing power parity (PPP) comparisons. PPP adjustments are based on differences in the average price level for all goods and services in an entire territory, measuring the cost of similar items in different places. This is done within countries using CPI item prices in the United States (BEA 2024) or similar methods elsewhere (Amendola et al. 2023), and done globally between all countries of the world by the International Comparison Program (World Bank 2024a).

2.1. Modern Data on Availability and Price of Commonly Consumed Items

The availability of data on which foods are sold at what price around the world was reviewed by Bai et al. (2021b), showing that by far the most complete and reliable coverage is provided by each country's CPI item lists designed to monitor general inflation and the internationally standardized PPP items used to compare price levels between countries. Almost every national government now computes a monthly CPI and reports prices for PPP comparisons every few years, with strong incentives for consistent reporting to maintain eligibility for international finance and aid flows. CPI and PPP data are intended to reflect nationally representative costs for a broad set of items, including dozens of each country's most commonly consumed foods. Other kinds of food price data have narrower coverage and less consistency over time, including market information and early warning system prices used to guide farm production and food distribution, or specialized sources tracking prices of specific items from particular marketing channels such as prices posted online, scanned from bar codes, or collected by individual researchers. Some diet cost studies use those specialized price data, but most use CPI and PPP prices that now underpin dozens of research papers analyzing seasonal, spatial, and other kinds of variation in diet costs and affordability (**https://sites.tufts.edu/foodpricesfornutrition**).

The CPI and PPP food items for which retail prices are routinely collected differ from country to country, and the individual item prices used to construct each CPI are not usually available to the public. Even when item prices are available, they may not be in a usable form, because the item description might not be sufficiently detailed to identify its food composition. Price data collection for CPI requires only that prices refer to the same item from month to month, with no need to specify its exact size or composition. In contrast, data collection for PPP requires that prices refer to the same item in multiple countries, leading to more explicit product descriptions reported as part of the International Comparison Program dataset. Also, unlike item prices used for national CPIs, the item prices used for international PPPs are available to researchers on request. They are compiled only every few years and provide only one national average price for each item, but due to their global coverage, high reliability, and official status as government statistics, these PPP prices are the foundation for global diet cost monitoring. Variation within countries is tracked with CPI prices and sometimes other sources, as reviewed by Bai et al. (2021b).

Country governments typically report price indexes based on a limited number of the most widely consumed items. In practice, as shown by Bai et al. (2022, figure S2), including more than about 100 items in the food list for PPP calculations has no impact on estimated least-cost diets, because the additional varieties included are usually higher-cost options in each food group. Only those countries that report fewer than about 100 food item prices are likely to omit some of the least expensive generic options needed for a healthy diet. Food lists used for diet cost calculations often omit hot prepared meals and ready-to-eat packaged foods, either because they are mixed items spanning several food groups, or because their nutritional composition is unknown. But even when they are included in a price list, prepared meals and ready-to-eat packaged items rarely have the lowest cost per unit of food, because their prices include value-added services and packaging. Food lists used for diet cost calculations also omit a household's own garden or other food production, gleaning or gathering and fishing, and omit any foods received as gifts or charity, because the purpose of calculating diet costs is to measure food access in retail markets.

Linking food prices to nutritional value begins with each item's food composition, sourced from food composition databases such as FoodData Central (USDA 2024) or those listed by the International Network of Food Data Systems (FAO 2024c) or European Food Information Resource Network (**https://www.eurofir.org**). Item sizes are initially reported in terms of weight, volume, or other natural units. The datasets describing each item's nutritional composition use the results of laboratory tests on foods and ingredients to report an item's edible portion and water weight, dietary energy, essential nutrients, and other compounds that affect health such as fiber. Accounting for edible portion and moisture content is especially important for vegetables and fruits as well as packaged and processed items that may be sold in dry or concentrated form or with high levels of water in the product. Composition data also allow classification of items into nutritionally defined food groups. For example, the high carbohydrate content of potatoes makes them a starchy staple in several nutritional classifications, while they are usually classified as a vegetable in production and market data. Nutritional classification schemes differ and evolve over time; for example, orange-fleshed sweet potatoes with high vitamin A content may be put in the same food group as carrots and other orange or red vegetables. Matching item descriptions to nutritional composition and food classification is among the most difficult steps in converting item prices to diet costs. To facilitate that step, the software tools for diet cost calculations provided by the Food Prices for Nutrition project include a food item information database listing more than 500 products commonly found in price datasets matched with their edible portions and total energy content per kilogram and classified by food groups used in dietary guidelines around the world (Food Prices for Nutrition 2024).

2.2. Affordability Relative to Income and the Evolving Definition of Food Security

Comparing diet costs to income could potentially be done in many ways, depending on the available data and the purpose of measurement. The current procedure used for global food security monitoring, done jointly by the Food and Agriculture Organization (FAO 2024a) and World Bank (2024b), uses each country's income distribution from recent household surveys, as compiled by World Bank (2023b), and subtracts nonfood spending observed on average around the national poverty lines adopted by governments at each level of income. As of October 2024, those international poverty lines are $2.15 per day in low-income countries, $3.65 in lower-middle-income countries, $6.85 in upper-middle-income countries, and $24.36 in high-income countries, of which the available household survey data show average nonfood spending by households around those income levels to be $0.80, $1.61, $3.70, and $13.20, respectively, each expressed in 2017 dollars at PPP prices (Bai et al. 2024, FAO et al. 2024, table S2.3). This approach was developed for the purpose of monitoring access to healthy diets in the context of other UN and World Bank metrics of global development, using national governments' choices about poverty lines to specify the minimum target for nonfood spending and replacing the food component of those national poverty lines with the least-cost locally available foods needed for health (Bai et al. 2024).

Before the availability of data and methods to compute least-cost diets as a measure of food access, the concept of food security—or its absence, food insecurity—was measured in a variety of ways, as discussed by Barrett (2010) and catalogued by Lele et al. (2016), among others. All of those were based on usual diets actually consumed by the population, relative to a desirable target. Global food security metrics of this type began with a method developed by P.V. Sukhatme (1961) to estimate the number of people in each country whose total food intake is below their estimated energy requirements. This approach, now known as the Prevalence of Undernourishment (PoU), is based on using survey data to estimate a log-normal distribution of energy intake among people, allowing shifts in each year's total population and national energy supplies to trace changes in the number and percentage of people with insufficient intake. Because the distributional parameters and estimated energy requirements used to measure PoU vary little from year to year, the changes are driven primarily by variation in total national supplies as measured by food balance sheets (FAO 2001).

The FAO's undernourishment metric was first used to track changes in global food security for all countries in the 1970s, in response to commodity price spikes and famines in several countries that led to the World Food Conference of 1974, where food security was defined as "availability at all times of adequate world food supplies of basic foodstuffs to sustain a steady expansion of food consumption and to offset fluctuations in production and prices" (United Nations 1975). During the 1980s and 1990s, increases in food production and total consumption drove down and stabilized market prices, which shifted attention to variation in how food is distributed within countries, and whether all people at all times are actually able to obtain the foods they need.

In the 1980s, a key methodological advance for measurement of food insecurity came from survey methods first developed at Cornell University by Kathy Radimer (1990). Questionnaires to elicit a person's experience of food insecurity, initially known as the Cornell-Radimer Hunger Scale, were first implemented nationally by the US Department of Agriculture (USDA) and the US Census Bureau in 1995. Modified versions remain widely used today, most commonly as the USDA Food Security Survey Modules (USDA ERS 2022) or the FAO Food Insecurity Experience Scale (FIES) questions developed under its Voices of the Hungry Project (FAO 2023c). Both the USDA and FAO surveys measure food insecurity using a set of questions about whether individuals, due to financial constraints or other limitations, have ever skipped a meal, ate less or differently, went to bed hungry, or were otherwise unable to obtain their usual diet at any time over the previous 12 months.

The FAO's undernourishment measure (PoU) refers to total energy derived from all foods, while food insecurity metrics (like FIES) refer to whether a respondent was ever without enough money or other resources to eat their usual diet. The USDA food security questionnaire asks specifically about whether the respondent ever, sometimes, or often could not afford "the kinds of food we want to eat," whereas the FAO questionnaire for FIES does not specify the type of food. This approach implies that food insecurity as measured by these questions is independent of the respondent's diet quality and level of food expenditure. The one question in the USDA household module about which foods are eaten asks whether there were times when respondents "couldn't afford to eat balanced meals," and the one question about diet quality for children asks whether they "relied on only a few kinds of low-cost food to feed our children" (USDA ERS 2022). The two corresponding questions in FIES relating to diet quality ask "whether, during the last twelve months, was there a time when, because of lack of money or other resources," the respondents "were unable to eat healthy and nutritious food" or "ate only a few kinds of foods" (FAO 2023c). These questions in USDA and FAO surveys represent a small step toward measuring access to nutritious items that would meet the 1996 definition of food security, asking people whether they could afford "balanced meals" in the USDA survey, or "healthy and nutritious food" in the FAO surveys, but those terms lack quantification and are left to the respondent to interpret. Use of these data to measure the respondent's access to nutritionally adequate or healthy diets is further obscured by reporting overall food insecurity as an aggregate score based on responses to multiple questions, such that responses to individual questions are rarely reported. Monitoring food security in a way that would capture access to healthy diets, as called for in the original 1996 mandate, requires other kinds of data and software tools to match food items with their nutritional characteristics, identify the lowest cost locally available options, and add up the resulting diet costs per day, as discussed below.

3. DEVELOPMENT AND USE OF BENCHMARK LEAST-COST DIETS FOR HEALTH

In economic terms, least-cost diets that meet nutritional requirements identify the frontier of cost-effectiveness for foods in a population's health production function (Grossman 1972). That counterfactual benchmark can then be contrasted with foods chosen based on revealed preferences in observed demand systems, as in Deaton & Muellbauer (1980). A variety of other goals and constraints might lead people not to consume the least-cost items needed for health, but even if people wanted and could afford a healthy diet, the required food attributes are unobservable without laboratory tests. Consumers cannot see, taste, or smell many of the most important components in food that will affect their future health, so buyers must rely on other information such as health claims and marketing materials (Masters et al. 2022). Because at least some consumers seek out and are willing to pay more for foods that would improve their health, sellers frequently offer items with health claims at premium prices, contributing to the widespread belief that healthier foods are generally more expensive (Haws et al. 2017). Studies using actual data on health-related attributes can compare costs per calorie for each type of food, as covered by Headey & Alderman (2019); per unit of an individual nutrient across all types of food, as discussed by Ryckman et al. (2021a,b; 2022); or per unit of a multidimensional quality index such as an item's Health Star Rating, Nutri-Score, or Food Compass Score, as pioneered by Martinez et al. (2024). These studies find that prices differ greatly by food group, with the lowest-cost sources of dietary energy being basic starchy staples, plant oils, and sugar. More nutrient-rich foods such as vegetables, fruits, legumes, nuts, and seeds or animal-source foods have higher retail prices, due to higher cost of production and distribution, but within each food group it turns out that healthier choices are not systematically more expensive. Some attributes needed for health add to the cost of food, but many attributes for which some consumers might pay a premium actually reduce the product's nutritional value, as measured by the Health Star Rating, Nutri-Score, or the Food Compass (Martinez et al. 2024).

A central challenge in comparing foods and measuring diet quality is that multiple attributes are all needed for survival and health, each complementing the others as joint inputs to biochemical and physiological processes inside the body (Masters & Finaret 2024). Overall diet quality scores such as those reviewed for the World Health Organization by Verger et al. (2023) take account of substitution among foods in delivering each attribute and also consider complementarity among the attributes needed to avoid deficiencies and protect against diet-related diseases. Systematic monitoring of access to all food attributes needed for a healthy diet was made possible only recently, based on matching CPI and PPP item prices to food composition and nutritional requirements per day.

Modern metrics of dietary requirements for health are typically represented in terms of fixed targets or upper and lower bounds for a set of specific attributes, based on laboratory, clinical, and epidemiological evidence about the normal range of variation within which people face relatively low disease risks. The role of food in health outcomes could potentially be modeled as a continuous function with dose-response based on relative risks, as in Wang et al. (2020), but the use of discrete upper and lower bounds is helpful for diagnostic purposes and also reflects the range of variation actually observed among healthy people.

Soon after the first essential nutrient requirements and food composition databases became available in the early twentieth century, Stigler (1945) described how an analyst might estimate the least-expensive combination of foods needed to meet recommended intake of each nutrient. The first linear programming algorithm to find an exact solution was then developed by George Dantzig (1963, 1990), in response to interest in least-cost diets and other optimization problems during and after World War II. Since then, least-cost diets for nutrient adequacy have been widely used to formulate rations, guide nutrition assistance, and constrain optimization to provide insight into actual food choices (Wallingford & Masters 2025).

In recent years, diet modeling has moved beyond essential nutrients to other requirements for a healthy diet specified in national food-based dietary guidelines. Modern dietary guidelines specify that a healthy diet should meet a person's energy requirements using a mix of food groups, as in the US MyPlate (USDA & HHS 2020) or UK Eatwell Guide (NHS 2024), and similar publications by dozens of governments worldwide, as compiled by the FAO (2024b) and reviewed by Herforth et al. (2019). Each country's guideline results from a government-sponsored effort to establish a scientific consensus on foods for health, as documented, for example, by Cullum

(2024). Analyses commonly start with how widely consumed foods can be combined so that total intake is likely to meet energy needs within the upper and lower bounds for all essential nutrients, and then consider various other food attributes influencing health. A diet model with nutritional constraints can be solved for a variety of objective functions, such as similarity to national average consumption (Wilde & Llobrera 2009), or goals based on palatability constraints (Gerdessen & de Vries 2015) and specific applications such as school meals (Stern et al. 2023), nutrition assistance (Wallingford et al. 2024), and other interventions (Knight et al. 2024).

Identifying foods for health when item prices are available is commonly done by solving for the least-expensive options, so as to identify the most cost-effective foods providing health attributes in the required proportions. These least-cost diet models begin with energy and nutrients, for example, in the formulation below:

$$\min_{cg}\left\{CostofNutrientAdequacy_{cg} = \Sigma_i \mathbf{p}_{\mathbf{ic}} \times q_{icg}\right\}, \tag{1}$$

subject to

$$\Sigma_i \mathbf{d}_{\mathbf{ie}} \times q_{icg} = \mathbf{EER}_{\mathbf{g}}, \tag{2}$$

$$\Sigma_i \mathbf{d}_{\mathbf{in}} \times q_{icg} \geq \mathbf{LB}_{\mathbf{ng}}, \tag{3}$$

$$\Sigma_i \mathbf{d}_{\mathbf{in}} \times q_{icg} \leq \mathbf{UB}_{\mathbf{ng}}, \tag{4}$$

$$q_{icg} \geq 0. \tag{5}$$

This illustrative model shown here would identify the most affordable foods to meet nutrient requirements in each country (*c*) for each demographic group (*g*); when added up over all food items (*i*) at local prices (**p**) in whatever quantities (*q*) would meet the group's estimated energy requirement (**EER**); and would keep each nutrient (**n**) above its lower bound (**LB**) and below its upper bound (**UB**) based on food composition data for the item's density (**d**) in terms of dietary energy (**e**) and individual nutrients (**n**). The items included in these benchmark diets differ by country, but the nutritional constraints they meet are drawn from the global evidence base such as the Dietary Reference Intakes for the United States and Canada produced by the National Academies (NASEM 2023) and the broadly similar Dietary Reference Values published by the European Food Safety Authority (Eur. Food Safety Auth. 2024). Drawing on the same global evidence base, combined with similarities but also differences in how foods provide nutrients, leads to systematic patterns in how foods can most cost-effectively be combined to meet diverse needs across and within national populations (Bai et al. 2022, Kuri et al. 2024).

Actual dietary guidelines start with nutrient adequacy and then add other health-related attributes for which each country's authorities identify a sufficient scientific consensus. Each national guideline development process draws on the same global evidence base, which often leads to similar conclusions, grouping items into nutritionally defined categories and recommending target levels of each food group. Herforth et al. (2022) used quantitative guidelines from 10 regionally representative countries and semiquantitative food guides from 30 countries to identify the most commonly recommended food groups and derive the average proportions recommended of each food group. These guidelines use a wide range of locally appropriate items and dishes to illustrate what a healthy diet looks like in their country, but they classify items into similar nutritionally defined groups with a similar recommended number and quantity of items from each food group.

The commonalities among national dietary guidelines identified by Herforth et al. (2022) are summarized in the Healthy Diet Basket (HDB), which specifies that a globally relevant benchmark healthy diet would include at least 11 locally available items from 6 globally standardized food groups. The HDB is calibrated to energy balance at 2,330 kcal/day, based on the needs of the median adult woman aged 19–30 in the World Health Organization's global reference population of healthy individuals, which happens to be almost exactly the average of all age–sex groups in that population. To meet nutrient needs and protect against diet-related diseases, the HDB specifies that the 11 items should include at least 2 starchy staples each at 580 kcal/day; one legume, nut, or seed product at 300 kcal/day; oil or fats at 300 kcal/day; two animal source foods each at 150 kcal/day; two fruits at 80 kcal/day each; and three vegetables at 37 kcal/day each. The target diversity in and between groups reflects the mix of nutrients provided by each type of food, for example, the high level of micronutrient density per kilocalorie of fruits and vegetables. As a result, using the least-cost options to meet these food group targets stays within the lower and upper bounds for almost all essential nutrients, and similar levels of nutrient adequacy are achieved with almost all countries' own national guidelines, as shown by Herforth et al. (2022, 2025).

The CoAHD developed by Herforth et al. (2020) has been reported annually for all countries in the *State of Food Security and Nutrition in the World* reports beginning in 2020 (FAO et al. 2020). Starting in 2022, the method was updated to use the HDB targets and adopted as an annual statistic published jointly by the FAO (2024a) and World Bank (2024b). Global monitoring of CoAHD is done using national average item prices from PPP comparisons over the entire year for each round of International Comparison Program (ICP) data collection. Country governments also track variation in access to that same global HDB target using monthly prices from each region collected for their CPI, as done since January 2024 by Nigeria's National Bureau of Statistics (Natl. Bur. Stat. 2024). Country governments can also use their CPI data to track access to a country-specific dietary guideline, as done, for example, in Ethiopia (Alemayehu et al. 2023, EPHI 2025), or draw on price data collected specifically for food market information systems, as in Ghana (Herforth et al. 2024).

In each case, identifying the least-cost items to meet food group targets is computationally and conceptually much simpler than diet modeling to meet nutrient requirements. Because food groups are mutually exclusive, least-cost diets to meet HDB targets require only rank-order comparison to select the lowest-priced items within each category, whereas diets for nutrient adequacy require linear programming to solve a system of simultaneous equations. The simplicity and clarity of meeting the HDB's targets have facilitated its adoption, along with availability of a software toolkit (Food Prices for Nutrition 2024) with Microsoft Excel templates as well as Stata and R code that is prepopulated with item descriptions matched to food composition data and nutritional targets. Those computational tools can then be installed on each analyst's own computer and integrated into local workflows to track healthy diet access in each country.

Prior to the development of HDB targets and the global CoAHD approach to tracking the cost and affordability of healthy diets, the more computationally difficult diet models based on linear programming were used for a wide range of purposes. The first published use of least-cost diets to track changing access to nutrient-adequate foods over time is O'Brien-Place & Tomek (1983) for the United States, followed much later by applications for low-income countries such as Uganda (Omiat & Shively 2017) and Malawi (Schneider 2022, Schneider et al. 2023). Comparisons among all countries of the world were pioneered by Allen (2017). Bai et al. (2021a) compare the minimum cost of purchasing a nutrient-adequate diet in 177 countries. For operational work within countries, the World Food Programme and its partners also developed specific Cost of the Diet software tailored to each population with whom they work (de Pee et al. 2017).

The development of new price indexes based on the lowest-cost items in each food group began by targeting the Minimum Dietary Diversity for Women (MDD-W) threshold, defined as consumption of at least 5 out of 10 nutritionally defined food groups (Arimond et al. 2010) that had been found to predict micronutrient adequacy (Martin-Prével et al. 2015). Tracking the lowest-cost items needed to include at least 5 of the 10 food groups each day provided a simple way to calculate percentage changes in the cost of meeting that dietary diversity target. This could then be compared to the earlier approach using linear programming to identify the lowest-cost items to achieve nutrient adequacy (Masters et al. 2018).

Since the development of the MDD-W, other metrics of diet quality use additional food group classifications, such as the Global Diet Quality Score that classifies intake as low, medium, or high levels of 25 distinct food groups (Bromage et al. 2021). The Global Dietary Recommendations score to limit risk and protect against noncommunicable diseases (NCDs) (comprising NCD-Risk and NCD-Protect scores) uses 18 distinct food groups (Herforth et al. 2020, WHO et al. 2024). Other diet quality metrics such as the US Healthy Eating Index (NIH 2024) use food group targets combined with other attributes such as whole grains (in contrast to refined grains) and unsaturated fats (in contrast to saturated fats). Just as CPIs evolve in response to changes in spending patterns (BLS 2024, Cavallo 2024), diet cost metrics can evolve to reflect the nutritional criteria used to define a healthy diet. Some of this work occurs in high-income countries, such as that by Penne & Goedemé (2021) for Europe, but interest in the lowest possible cost of a healthy diet arises primarily in low- and middle-income countries, for which a recent summary of diet cost monitoring efforts is provided by Herforth et al. (2024).

To illustrate the growth and composition of least-cost diet research, we conducted a nonexhaustive but systematic search for English-language studies using cost minimization to identify food items and quantities for a human diet satisfying a set of nutritional or food group requirements. An initial search of Google Scholar led to 363 candidate studies, from which we excluded conference abstracts, dissertations, non-English-language items, and research articles that did not model human diets, leading to 42 items retained for full-text screening. Of those, 26 studies actually used cost minimization and were retained, and full-text screening also revealed citations to an additional 28 articles that met our inclusion criteria despite not having our search terms in their title. The result of this search is a set of 54 studies published between 1945 and 2023, which we classified by type of nutritional constraints, as shown in **Figure 1**.

Figure 1 reveals that after Stigler (1945), there were only occasional published studies using least-cost diets for nutrient adequacy until they began to appear almost every year from 2012 onward. Then studies appeared in 2016 and 2018 that introduced food group constraints as well as nutrient requirements. That sequence began with a study in Copenhagen to quantify the additional costs beyond nutrient adequacy of meeting a variety of other constraints (Parlesak et al. 2016). This was followed by a study in Ghana and Tanzania of monthly and regional variation in the lowest-cost items to meet MDD-W standards of diet diversity and also nutrient requirements (Masters et al. 2018) and by studies in South and Southeast Asia tracking the cost of meeting local dietary guidelines (Dizon et al. 2019, Mahrt et al. 2019). Those national studies were soon followed by global analyses using item prices reported for PPP calculations, first to meet national governments' dietary guidelines (Herforth et al. 2020) and EAT-*Lancet* recommendations (Hirvonen et al. 2020), then the HDB standard (Herforth et al. 2022), as presented below.

4. RESULTS OF GLOBAL LEAST-COST DIETS IN CONTRAST TO ACTUAL FOOD SPENDING

The first and most important finding from using least-cost healthy diets to measure food access globally is summarized in **Figure 2**, presenting benchmark results computed from item prices for 168 and 173 countries in 2017 and 2021, along with actual food expenditure from national accounts for 168 and 166 countries in 2017 and 2021, respectively. Both are reported by the International Comparison Program, from which diet cost and food expenditure results are published jointly by the FAO and World Bank.

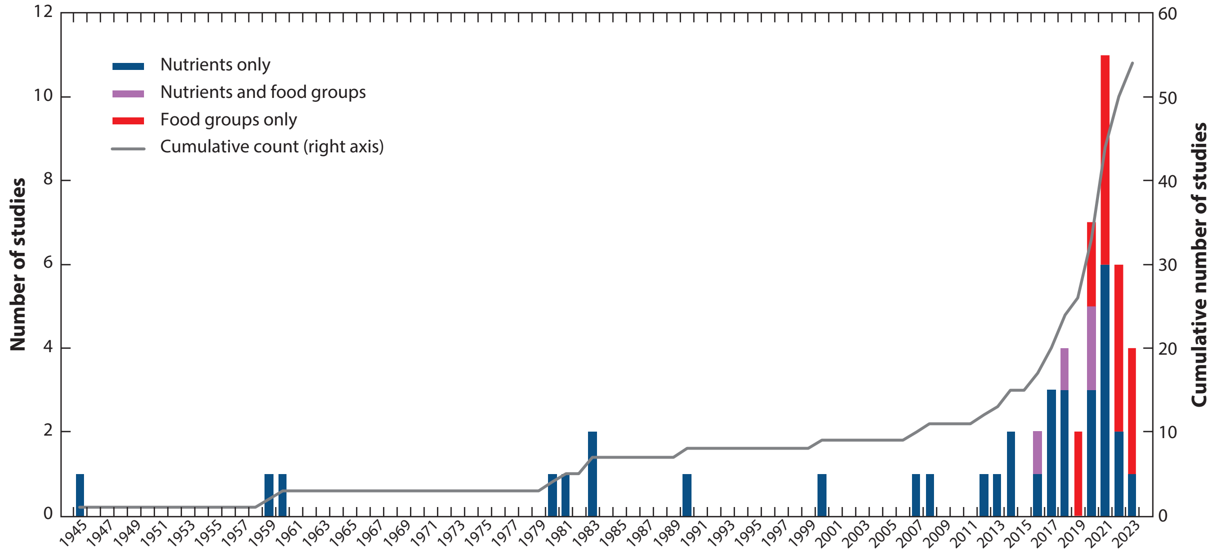

Figure 1

Annual and cumulative number of least-cost diet studies by type of constraint, 1945–2023. Data shown are the number of least-cost diet studies identified through Google Scholar and PubMed and references cited by later studies. Bibliographic data on each study are shown in **Supplemental Table 1**, accompanied by a PRISMA diagram showing the full count of studies found (**Supplemental Figure 1**).

Supplemental Material >

The global pattern in **Figure 2** reveals how the least-expensive items in quantities needed for a healthy diet typically cost about \$3–4/day in constant US dollars at PPP prices. That cost level is not correlated with national income. In contrast, actual food spending rises with income, from far below the minimum cost of healthy foods in low-income countries, to far above that minimum cost in high-income countries. These diet costs are based on food item prices used for PPP comparisons, and food spending is from national accounts. Both kinds of data are collected by government statistical organizations and intended to be nationally representative, reported through the International Comparison Program (World Bank 2024a). The series of diet costs from 2017 onward is available from FAO (2024a) and World Bank (2024b).

The central discovery illustrated by **Figure 2** is that least-cost healthy diets generally cost between \$2.50 and \$5.00 per day. National average food spending ranges from around \$1 to more than \$10 per day, and national income varies from less than \$5 to more than \$100 per day. Analyses within countries reveal the same pattern, with seasonal or spatial variation and periodic food price spikes causing minimum diet costs to fluctuate in real terms by a factor of about two. Actual food spending can vary by a factor of ten or more, and real incomes within countries can vary much more than that. Analyses of least-cost diets are just beginning to reveal the kind of spatial and seasonal patterns first documented across East Africa by Bai et al. (2020), partly through the accumulation of more studies with existing methods and also through a variety of methodological improvements, as suggested, for example, by Headey et al. (2024b).

5. POLICY USES AND FUTURE DIRECTIONS FOR LEAST-COST DIETS RESEARCH

The least-cost diet research described in this review has already revealed important insights for policies and programs around the world, leading to a variety of recent initiatives and potential innovations to provide even more useful guidance.

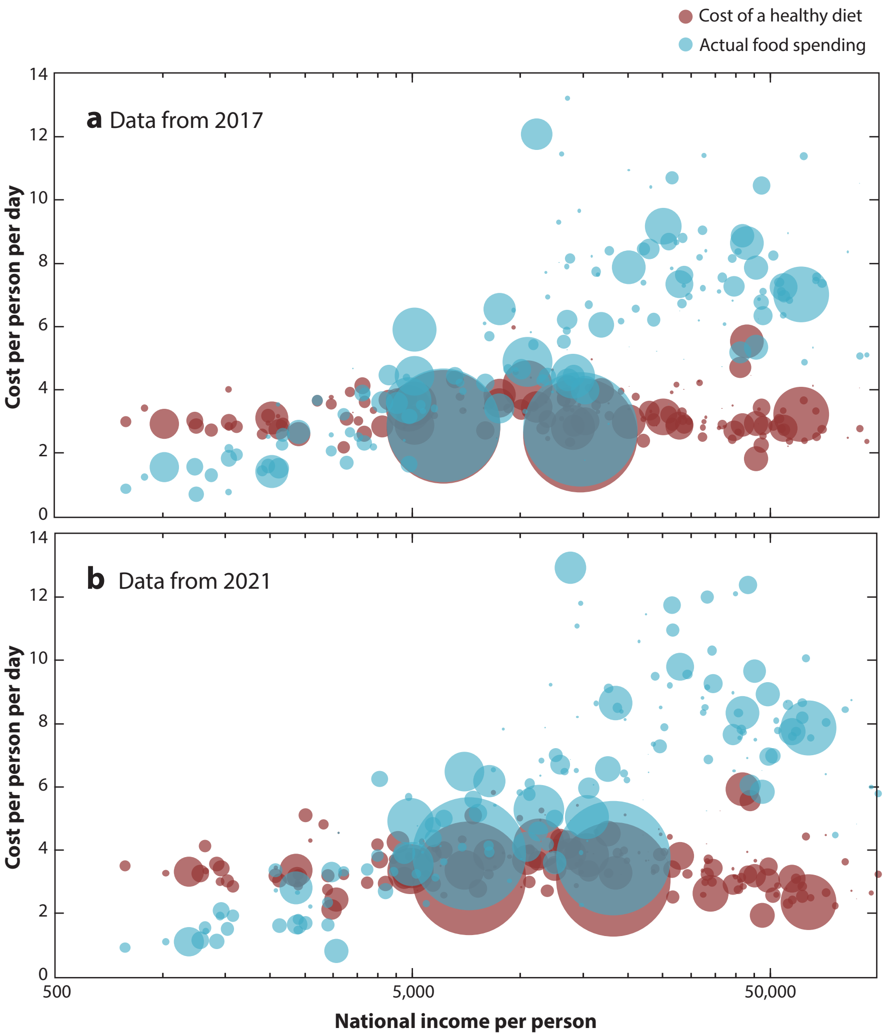

Figure 2

Least-cost healthy diets and actual food spending by national income: (*a*) 2017 and (*b*) 2021. Data shown are FAO and World Bank estimates, all in 2017 US dollars at purchasing power parity (PPP) prices. Diet costs use item prices from national statistical organizations, and food expenditure is from national accounts, both reported through the International Comparison Program (ICP). The most recent diet cost data are always available from FAO (2024a), while archived versions are at World Bank (2024b), including data shown here, which are Food Prices for Nutrition dataset 1.0 for 2017 (diet costs July 2022) and dataset 3.0 for 2021 (released July 2024). National income (GNI) per person at PPP prices in 2021 is reported in current US dollars, deflated to 2017 US dollars using the US consumer price index from the World Bank (2024c).

A first use of least-cost diets in food policy analysis is to identify the frontier of cost-effectiveness in delivering foods for health, showing which items can deliver each food group at the lowest cost per day, and revealing where or when consumers are not being reached by those low-cost supply chains. For example, the relatively high cost of healthy diets found in Latin America and the Caribbean has already triggered analysis and advocacy to improve production and distribution of lower-priced options in required food groups (FAO et al. 2023), and some initial work has studied whether and how import restrictions affect access to low-cost options for a healthy diet (Gilbert et al. 2024). Older food price metrics focus on farmgate prices and agricultural commodity markets, such as FAO (2023a). However, the retail cost of food paid by households mostly pays for postharvest storage and distribution to the point of sale, in contrast to farmgate or commodity prices that typically account for less than one-third of retail food costs even in low- and middle-income countries (Yi et al. 2021, FAO 2023b).

A second important use is to track affordability of even the least-expensive options relative to incomes available for food, revealing the need for higher earnings, social protection, and nutrition assistance. One early example of impact on social protection concerns debates over the appropriate minimum wage in Nigeria, where news accounts report that the National Bureau of Statistics' diet cost data were helpful in anchoring negotiations between trade unions and the government (Gulloma 2024).

A third application is to compare benchmark diets to foods actually chosen, so as to identify where and when least-cost items in a healthy diet are displaced by more expensive and often less-healthful options. Food choice among affordable options is driven by a wide range of potentially modifiable factors, including the ways that food marketing might alter consumers' aspirations as well as taste and cultural factors that drive choices toward or away from the products needed for health (Costlow et al. 2025). Public funds are sometimes used to help consumers meet health needs at low costs, such as the Shop Simple with MyPlate app for low-cost recipes that meet the Dietary Guidelines for Americans (**https://www.myplate.gov/app/shopsimple**).

Beyond the three kinds of work with existing methods described above, methodological innovations could provide more timely and accurate analysis for decision-makers around the world. Four kinds of new techniques to improve measurement of diet costs are described below.

One set of innovations concerns food price data collection, where improvements could address both systematic biases and random errors related to the selection of food items, market locations, timing of data collection, and other factors. Bias could arise because prices collected to measure inflation aim to capture total expenditure including premium items, whereas measuring food access calls for data on the least-expensive options in marketplaces serving populations at risk of malnutrition and diet-related disease. Beyond those potential biases, there are opportunities to improve accuracy and timeliness by upgrading data collection and analysis from older paper-based and other error-prone systems to modern systems that use preprogrammed tablets for nearly immediate feedback to field staff about data quality and nearly immediate forwarding of results for processing into diet costs and affordability results. One recent example concerns web scraping online prices to track availability and price of the lowest-price options by the UK Office of National Statistics (ONS 2022) and in research (Cavallo & Kryvtsov 2024).

A second type of innovation concerns methods for data analysis and transformation into diet costs. The first step of the process, matching items to food composition, could be improved with new data sources and matching tools, including estimates of variance and changes in the energy density and nutrient composition of each food. The second step of specifying dietary requirements is continuously improving, thanks to innovations in dietary guidelines, definitions of diet quality, and food ratings. Each kind of data has variance that could potentially be aggregated into estimated confidence intervals for each diet cost, including uncertainty about prices and also food composition or dietary requirements as in Bai et al. (2022).

A third focus of innovation could be to expand the agenda beyond access and affordability at the market to consider meal preparation costs as a factor in food choice and a barrier to consumption of healthy diets. Existing work focuses on market prices and diet costs for retail items, but being able to afford the ingredients for a meal is just the first step toward improved nutrition. Meal preparation practices and associated costs are highly variable across households, and there is not yet any practical way to measure the cost differentials in meal preparation between different foods and dishes that might comprise a healthy diet in each context. However, new methods are being developed as researchers and decision-makers, including the Indian government, shift their focus from individual foods to meals and daily diets (Masters et al. 2021, Ministry of Finance 2020).

Finally, there are many opportunities for the data and methods developed in monitoring healthy diet access to be used for development goals beyond nutrition and health. Much of the same data about retail markets and food use can be used to consider the cost and affordability of reaching goals for environmental sustainability, social inclusion, and other concerns. Those questions can be addressed using modeled diets or for individual foods using a true-cost accounting framework (Kennedy et al. 2023).

6. CONCLUSIONS

In summary, a long tradition of using food prices and least-cost diets to understand access to food has led to monitoring of the cost and affordability of diets as a global indicator of food security, with important insights for improving the availability and affordability of foods in the quantities needed for health. Historical retail prices, as well as analyses of commodity prices and national accounts data, provide insights into the volatility and composition of retail prices to inform their use in monitoring physical and economic access to healthy diets. Efforts from the Food Prices for Nutrition project and international institutions to monitor the cost and affordability of healthy diets provide policy-relevant diagnostics where healthy diets are unaffordable and where millions who can afford the monetary cost of healthy diets actually eat other foods instead. These insights highlight important areas for future research, including options for improving the precision of estimates and incorporating nonmarket or hidden costs that affect consumption decisions, such as meal preparation or environmental costs. High-frequency, subnational monitoring using existing retail food price data collected by national governments can support decision-making and methodological advances.

DISCLOSURE STATEMENT

The authors are not aware of any affiliations, memberships, funding, or financial holdings that might be perceived as affecting the objectivity of this review.

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